



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Multimedia representation and perception

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Multimedia Communications



Compression principles



Introduction

- ▶ Multimedia traffic dominates today's internet: Video streaming alone accounts for over 80% of all internet traffic
- ▶ This course provides practical skills for the digital age: learn to design and implement multimedia systems and understand real-world protocols and standards
- ▶ Evolution of multimedia technologies: from established standards (JPEG, MPEG, H.26x) to emerging trends in neural-enhanced coding and semantic delivery
- ▶ Interdisciplinary approach combining elements of signal processing, information theory, and neural networks
- ▶ Hands-on experience:
 - ▶ Practical labs simulating real-world scenarios and system implementation
 - ▶ Usage of industry-standard tools and introduction to modern AI frameworks
- ▶ Foundation for advanced studies: preparation for specialized courses in video coding, network optimization, and research opportunities (MSc/PhD)



Outline

The human visual system

The sound perception

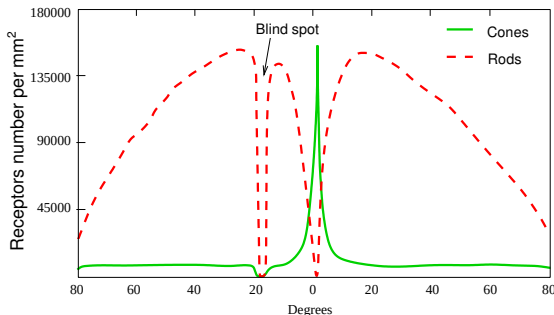
Image and video representation

Compression principles



The eye

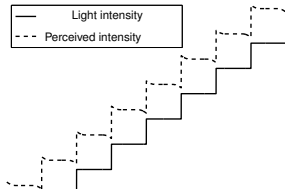
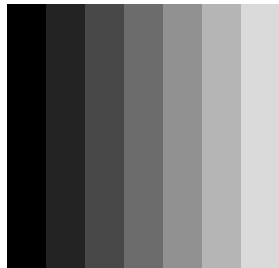
- ▶ Light is transformed into neural impulses by the retinal receptors
 - ▶ **Cones** (6÷7 millions, center of the retina) : very sensitive to colors, good resolution, need high illumination
 - ▶ **Rods** (75÷150 millions): sensitive to light intensity low resolution, very sensitive in low illumination





Light perception

- ▶ Perceived intensity : logarithmic function of intensity
- ▶ Intensity dynamic range: $\approx 10^{10}$ (100dB)
- ▶ The human visual system (HVS) cannot operate into such a large range at once
- ▶ Global lighting changesm reduced range dynamics
- ▶ Perceived light intensity: it is not just a function of light intensity





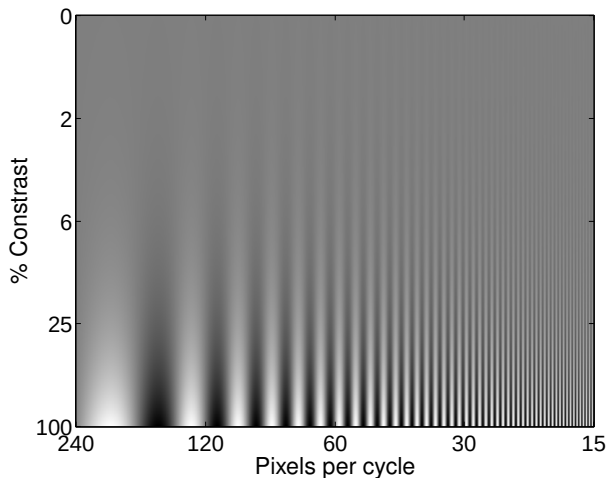
Contrast sensitivity function (CSF)

The Contrast Sensitivity Function (CSF) is a measure of the human visual system's ability to detect differences in luminance across various spatial frequencies. It quantifies the minimum contrast required for an observer to perceive a pattern at different scales of detail.

- ▶ **Contrast:** This refers to the difference in luminance between the lightest and darkest parts of a pattern. It is expressed as a percentage
- ▶ **Spatial frequency:** This is measured in **cycles per degree** of visual angle and represents how often a pattern repeats over a given distance. Low spatial frequencies correspond to large-scale patterns, while high spatial frequencies represent fine details
- ▶ **Sensitivity:** the reciprocal of the minimum contrast required to detect a pattern. Higher sensitivity means the ability to detect lower contrasts.



Contrast sensitivity function (CSF)

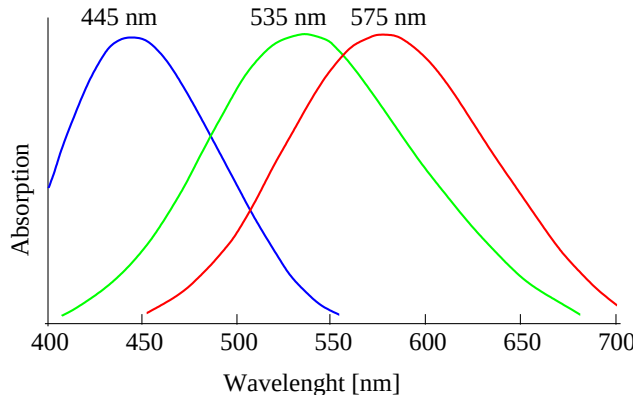


The CSF is typically determined through psychophysical experiments where observers are presented with sine-wave gratings of varying spatial frequencies and contrasts. The threshold contrast at which the observer can just barely detect the pattern is recorded for each spatial frequency. The CSF usually peaks at mid-range frequencies (around 2-5 cycles per degree) and declines at both lower and higher frequencies.



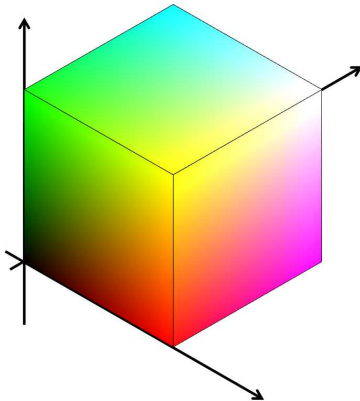
Color Perception

- ▶ Visible light spectrum: $400 \div 700$ nm
- ▶ Cones sensitivity:
 - ▶ 65% sensitive to red
 - ▶ 33% sensitive to green
 - ▶ 2% sensitive to blue (but very sensitive)
- ▶ Color sensation: corresponds to the *tristimulus*
- ▶ Color is obtained as a combination of the three primary colors

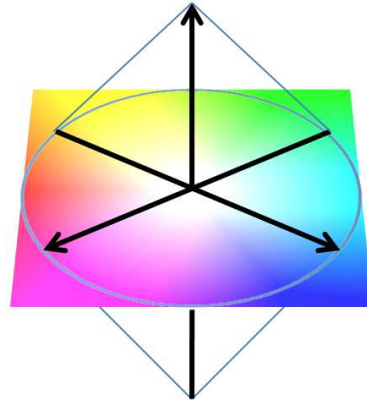




Color spaces



RGB



HSV



Color spaces

The RGB color model is represented as a unit cube:

- ▶ Origin (0,0,0): Black
- ▶ Opposite corner (1,1,1): White
- ▶ Main diagonal: Grayscale
- ▶ Primary colors at (1,0,0), (0,1,0), (0,0,1)
- ▶ Secondary colors at (0,1,1), (1,0,1), (1,1,0)

Transformation to HSV:

- ▶ Balance RGB cube on its black corner
- ▶ Rotate so black-white diagonal is vertical
- ▶ View from above (appears as hexagon)

In HSV representation:

- ▶ Hue: Angle around vertical axis (0° - 360°) selects color family
- ▶ Saturation: Distance from central axis, determines purity
- ▶ Value: Vertical position (0 at bottom, 1 at top), controls brightness



Color spaces

- ▶ One of the most relevant color spaces is called YCbCr
- ▶ It is very similar to HSV
- ▶ The Y component is the same as HSV value (ie., luminance)
- ▶ Cb and Cr components convey the chrominance information (just as Saturation and Value)
- ▶ While S and V represent chrominance in polar coordinate, Cb and Cr represent it in Cartesian coordinates
- ▶ Very often, the YCbCr space is referred to as YUV (U and V are the chrominance components)



Mathematical Representation of Color Conversion

- ▶ A pixel in the RGB space is represented as a vector: $\mathbf{p}_{RGB} = [R, G, B]^T \in [0, 1]^3$.
- ▶ The transformation to the YCbCr space is a linear mapping (ITU-R BT.601):

$$\begin{bmatrix} Y \\ Cb \\ Cr \end{bmatrix} = \begin{bmatrix} 0.299 & 0.587 & 0.114 \\ -0.1687 & -0.3313 & 0.5 \\ 0.5 & -0.4187 & -0.0813 \end{bmatrix} \begin{bmatrix} R \\ G \\ B \end{bmatrix} + \begin{bmatrix} 0 \\ 128 \\ 128 \end{bmatrix} \quad (1)$$

- ▶ **Luminance (Y)**: Captures the "brightness" using weights that reflect the HVS sensitivity (maximum for Green, minimum for Blue)
- ▶ **Chrominance (Cb, Cr)**: Represent the color differences ($B - Y$ and $R - Y$). The offset (128 for 8-bit systems) is used to center the range
- ▶ **Matrix Inversion**: Since the transformation is non-singular, we can perfectly reconstruct RGB from YCbCr: $\mathbf{p}_{RGB} = \mathbf{M}^{-1}(\mathbf{p}_{YCbCr} - \mathbf{o})$.



Beyond Human Perception: Machine-Centric Multimedia

- ▶ **A New Type of Consumer:** A significant portion of multimedia traffic is now consumed by **algorithms** (e.g., automated surveillance, industrial monitoring, IoT) rather than humans
- ▶ **Human vs. Machine Perception:**
 - ▶ **Human Visual System (HVS):** Prioritizes semantic regions like faces and is limited by Contrast Sensitivity (CSF)
 - ▶ **Machine Models:** Often rely on high-frequency details and local textures that the HVS might ignore, but are crucial for **feature extraction**.
- ▶ **Task-Oriented Communications:**
 - ▶ Traditional goal: Minimize perceived distortion
 - ▶ Modern goal: Maximize the **accuracy of a downstream task** (e.g., "Was the object correctly identified?") at the minimum possible rate
- ▶ **Compression Implication:** If the end-user is a machine, we can discard information that is visually important but "semantically" irrelevant for the specific task, enabling extreme compression ratios



Compression principles



Pure tone perception

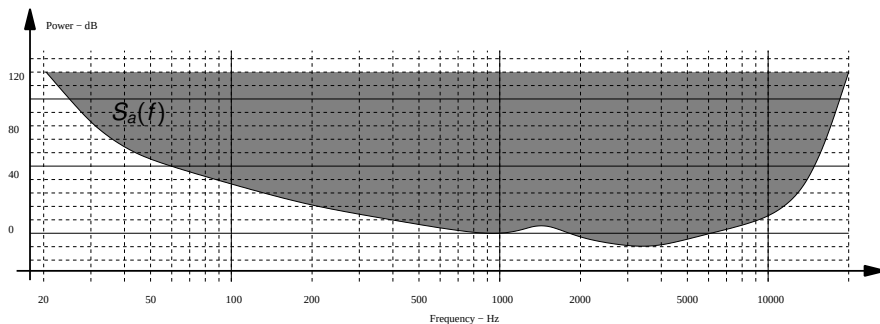
- ▶ Pure tone : $x(t) = a \sin(2\pi f_1 t)$ sinusoid with power $\sigma^2 = \frac{a^2}{2}$
- ▶ This sound excites several nerves (power spreading)
- ▶ Model : filterbank with M filters
 - ▶ The k -th filter corresponds to the k -th nervous fiber
 - ▶ Frequency response of the k -th filter: $H_k(f) = A_k(f) \exp^{j\phi_k(f)}$
 - ▶ Responds to the f_1 sinusoid:

$$y_k(t) = aA_k(f_1) \sin [2\pi f_1 t + \phi_k(f_1)]$$

- ▶ The power ratio is called spreading function: $S_E(k) = A_k^2(f_1)$



Hearing threshold

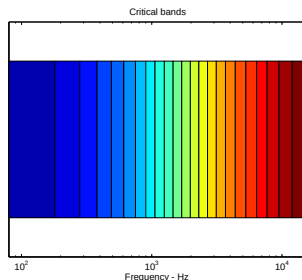


- ▶ The minimum power minimale for a tone at frequency f to be audible is $S_a(f)$
- ▶ $S_a(f)$ is a function of f and has a minimum between 1 and 4kHz (speech)



Critical band (CB)

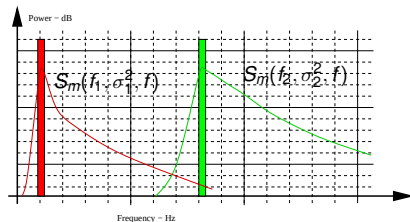
- ▶ A pure tone at frequency f_1 must have a minimum power $\sigma_1^2 > S_a(f_1)$ to be heard
- ▶ For N sinusoids *near* f_1 we need only $\sum_i \sigma_i^2 > S_a(f_1)$
- ▶ Sinusoids are *near* if they are in the *critical band*
- ▶ CB amplitude is a function of f_1







Frequency masking functions $S_m(f_0, \sigma^2, f)$



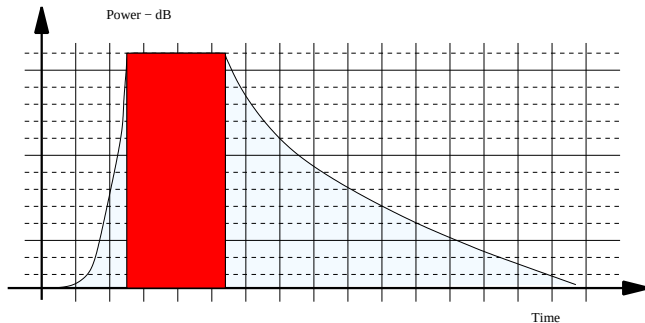
- ▶ For given f_0 and σ^2 , $S_m(f)$ has a triangular shape
- ▶ The maximum is achieved at $f = f_0$
- ▶ Masking index: $S_m(f, \sigma^2, f) - \sigma^2$

- ▶ we observe that the second sound has not necessarily to be more powerful than the first to be heard: $S_m(f, \sigma^2, f) < \sigma^2$
- ▶ Decrease is slower for increasing f_1
- ▶ Decrease slope is proportional to CB
- ▶ Slope (toward increasing frequencies) is a decreasing function of σ^2



Masking curves

Time masking



- ▶ Pre-Masking : 2÷5 ms
- ▶ Post-Masking : 100÷200 ms



Model application

- ▶ The psychoacustical model allows to find out some non-audible parts of the signal
- ▶ We may then allow some quantization noise, as soon as it is masked by the rest of the signal
- ▶ Nevertheless, the model is not perfect:
 - ▶ Only pure tones or narrowband sounds are considered
 - ▶ We are only able to assess the influence of 3 sounds at a time
 - ▶ Real-life signals are much more complex: how do they interact?
- ▶ In practice, compression algorithms parameters are determined in an experimental way, after a large number of tests



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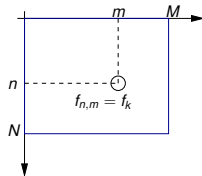
Image and video representation

Compression principles



Gray level images

- ▶ Discrete grid, $N \times M$ pixels
- ▶ A given position (m, n) is scanned in raster order k
- ▶ $k = (n - 1)M + m$
- ▶ $f_{n,m} = f_k$





RGB representation

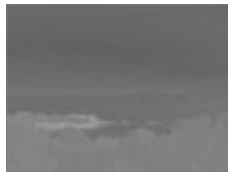
Color images have three components, each represented as a gray scale image.





YUV representation

Color images: one luminance component and two chrominance components





Color Sampling

Sampling schemes

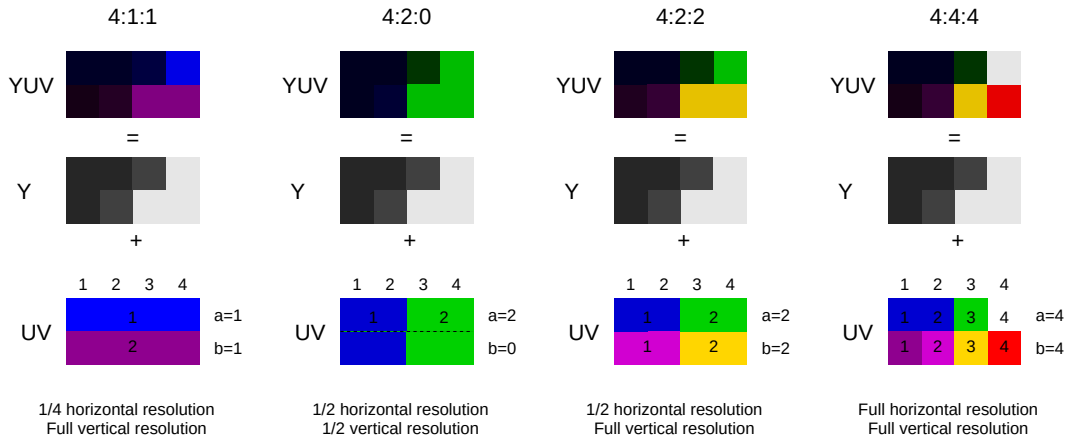
- ▶ The sampling scheme is represented with three integers

$$J : a : b$$

- J** Reference horizontal size, usually equal to 4
- a** Number of chroma sample on the first line of the reference pattern
- b** Number of further samples on the second line of the reference pattern



Color sampling





YCbCr 4:2:0

YCbCr 4:2:0 is a color space encoding scheme that leverages human visual perception to achieve image compression. It is the digital version of YUV. It separates luminance from chrominance information, taking advantage of the human eye's higher sensitivity to brightness changes compared to color changes.

Key components:

- ▶ **Y**: Luminance (brightness) component
- ▶ **Cb**: Blue-difference chroma component
- ▶ **Cr**: Red-difference chroma component

This format is widely used in digital video and image compression standards due to its efficiency in reducing data while maintaining perceived quality.



YCbCr 4:2:0

Compression Mechanism:

- ▶ In RGB all channels are sampled at full resolution
- ▶ In YCbCr 4:2:0
 - ▶ Luminance (Y) is sampled at full resolution
 - ▶ Chrominance (Cb, Cr) is subsampled by a factor of 2 both horizontally and vertically

Compression Ratio: $\frac{4+1+1}{4+4+4} = \frac{6}{12} = 0.5$

This results in a 50% data reduction compared to full RGB sampling.

Perceptual Basis:

- ▶ Human vision is more sensitive to brightness changes than color changes
- ▶ Full Y resolution maintains perceived sharpness
- ▶ Reduced Cb, Cr resolution aligns with human color processing capabilities
- ▶ High perceived quality is maintained despite significant data reduction



Quantization

- ▶ Samples represented on a discrete set
- ▶ Uniform quantization (rounding)
- ▶ L = number of levels
- ▶ $b = \log_2 L$ quantizer dynamics
- ▶ Typically $b = 8$ per component
 - ▶ 256 gray levels (8 bpp)
 - ▶ 16M colors (24 bpp)
- ▶ High dynamics range : 32 to 64 bits per channel



Resolution

<i>SECAM</i>	384 × 576	50 Hz
<i>PAL</i>	450 × 576	50 Hz
<i>NTSC</i>	323 × 486	60 Hz
QCIF	144 × 176	N/A
CIF	288 × 352	N/A
4CIF	576 × 704	N/A
SD/PAL	720 × 576	50 Hz
HD 720p/i	1280 × 720	50/100 Hz
HD 1080p/i	1920 × 1080	50/100 Hz
4K UHD	3840 × 2160	up to 120 Hz
8K UHD	7680 × 4320	up to 120 Hz



Representation of digital video

- ▶ Sequence of digital images
- ▶ Time dependency
- ▶ Three color components
- ▶ RGB or YUV representation
- ▶ Subsampling of color components

$$I : (n, m, T, c) \rightarrow x \in \{0, 1, \dots, 2^b - 1\}$$



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Compression: Motivations

- ▶ HD DVB System
 - 1 luminance component 1920×1080
 - 2 chrominance components 960×540
 - 8 bits quantization
 - 50 images per second
 - What is the rate in bits per second?
 - How much space is needed for a 2-hours video?



Compression: Motivations

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 - 1 luminance component 1920×1080
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 - 8 bits quantization
 - 50 images per second
 - What is the rate in bits per second?
 - How much space is needed for a 2-hours video?
 - $R \approx 1.24$ Gbps
- ▶ 2-hours movie ≈ 1.12 TB



Compression fundamentals

Why is it possible to compress?

- ▶ **Statistical redundancy:**
 - ▶ Spatial: images are often locally homogeneous
 - ▶ Temporal: successive images are highly correlated
- ▶ **Psychovisual redundancy:**
 - ▶ Limits of the HVS (CSF, Masking, Contours)
 - ▶ Goal: "Visually lossless" coding
- ▶ **New: Semantic redundancy:**
 - ▶ Concepts and objects: images contain recognizable structures.
 - ▶ Goal: "Generative" coding (transmitting tokens/meaning instead of pixels)

Modern algorithms should exploit all three to maximize performance



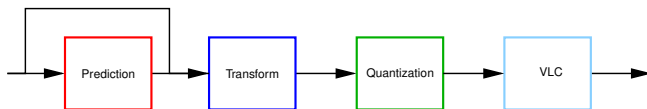
Lossless and lossy algorithms

- ▶ Lossless algorithms
 - ▶ Perfect reconstruction
 - ▶ Based on statistics
 - ▶ Small compression ratio
- ▶ Lossy algorithms
 - ▶ Decoded $\neq$ original
 - ▶ Based on quantization
 - ▶ Psychovisual redundancy: "visually lossless"
 - ▶ High compression ratio



Basic tools for compression

- ▶ **Transform:** Linear (DCT, Wavelet) or **Non-linear (Neural)** transforms to concentrate information
- ▶ **Prediction:** Alternative method for redundancy reduction (spatial or temporal)
- ▶ **Quantization:** Rate reduction through approximation (the only lossy step!)
- ▶ **Entropy Coding:** Residual redundancy reduction (VLC, Arithmetic)





Rate

Compression ratio

$$\blacktriangleright T = \frac{B_{\text{in}}}{B_{\text{out}}} = \frac{R_{\text{in}}}{R_{\text{out}}}$$

Coding rate

$$\blacktriangleright \text{Image : } R = \frac{B_{\text{out}}}{NM} \text{ [bpp]}$$

$$\blacktriangleright \text{Video, audio : } R = \frac{B_{\text{out}}}{T} \text{ [bps]}$$

Lossless image coding: $T \leq 3$

Lossy image coding: $T \approx 5 \rightarrow ?$

Lossy video coding: $T \approx 20 \rightarrow ?$



Quality and distortion

Criteria for image quality evaluation

- ▶ **Objective criteria** are mathematical functions of:
 - ▶ $f_{n,m}$: original; and
 - ▶ $\tilde{f}_{n,m}$: decoded image
- ▶ Non-perceptual objective criteria
 - ▶ Do not take into account the HVS characteristics
- ▶ Perceptual objective criteria
 - ▶ Based on perception models



Criteria for image quality evaluation

Non-perceptual objective criteria (NP-OC)

- ▶ Error image: $\mathcal{E}(f, \tilde{f}) = f - \tilde{f}$
- ▶ **Mean Square Error (MSE)** $\mathcal{D}$:

$$\mathcal{D}(f, \tilde{f}) = \frac{1}{NM} \|\mathcal{E}\|^2 = \frac{1}{NM} \sum_{n=1}^N \sum_{m=1}^M \mathcal{E}_{n,m}^2$$

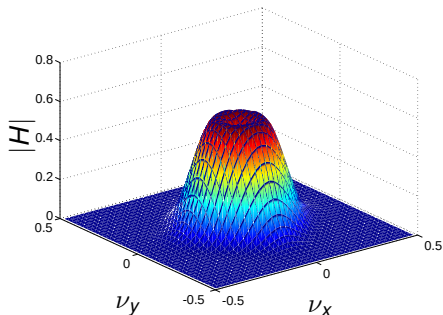
- ▶ **Peak signal-to-noise ratio** : $\text{PSNR}(f, \tilde{f}) = 10 \log_{10} \left(\frac{255^2}{\mathcal{D}(f, \tilde{f})} \right)$
- ▶ Simple, derivable, linked to the $\mathcal{L}^2$ norm



Criteria for image quality evaluation

Perceptual objective criteria

Weighted PSNR : Given a frequency weighting function (filter) h :



$$\text{WPSNR}(f, \tilde{f}) = 10 \log_{10} \left(\frac{255^2}{\mathcal{D}_W(f, \tilde{f})} \right) \quad \text{where} \quad \mathcal{D}_W(f, \tilde{f}) = \frac{1}{NM} \|h * \mathcal{E}\|^2$$



Criteria for image quality evaluation

Perceptual objective criteria

Structural Similarity Index (SSIM Index) between two blocks x et y :

$$\text{SSIM}(x, y) = [l(x, y)]^\alpha \cdot [c(x, y)]^\beta \cdot [s(x, y)]^\gamma$$

$$l(x, y) = \frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1} \quad \text{Luminance}$$

$$c(x, y) = \frac{2\sigma_x\sigma_y + C_2}{\sigma_x^2 + \sigma_y^2 + C_2} \quad \text{Contrast}$$

$$s(x, y) = \frac{\sigma_{xy} + C_3}{\sigma_x\sigma_y + C_3} \quad \text{Structure}$$

For the sake of simplicity, $\alpha = \beta = \gamma = 1$, $C_3 = C_2/2$

$$\text{SSIM} = \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)}$$

SSIM between images is the average of blocks' SSIM



NN-based Quality Assessment

- ▶ **Limits of SSIM/PSNR:** They often fail to capture "realism" in high-compression or AI-generated content
- ▶ **Deep Perceptual Metrics:**
 - ▶ Use pre-trained Neural Networks (e.g., VGG, ViT) as feature extractors.
 - ▶ Measure distance in the "feature space" rather than pixel space.
 - ▶ Example: **LPIPS** (Learned Perceptual Image Patch Similarity).
- ▶ **Key Concept:** A signal can have low PSNR but high perceptual "plausibility".



Criteria for image quality evaluation

Subjective criteria (SC)

- ▶ Subjective criteria are based on the image assessment performed by human observers
 - ▶ Difficulty of HVS modelling for objective criteria
 - ▶ Statistical analysis of results
 - ▶ Long, difficult and costly evaluations
- ▶ Often NP-OC are used
 - ▶ Simplicity
 - ▶ Geometrical interpretation (norm)
 - ▶ Analytical Optimization
 - ▶ Correlation with SC?

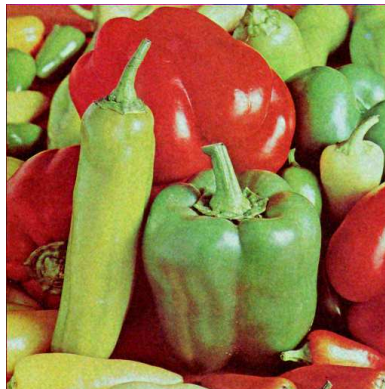


Error perception

Distributed error, white noise $\sigma = 4$



MSE: 16



SSIM: 0.906



Error perception

Error concentrated over 100×100 pixels



MSE: 16



SSIM: 0.972



A close-up photograph of a variety of colorful bell peppers. In the center, a large green bell pepper is prominent. To its left is a long, slender yellow-green pepper. Above the green pepper is a large, vibrant red bell pepper. To the right of the green pepper is a yellow bell pepper. The background is filled with more peppers in various colors and shapes, creating a rich, textured scene. The lighting is bright, highlighting the glossy surfaces of the peppers.

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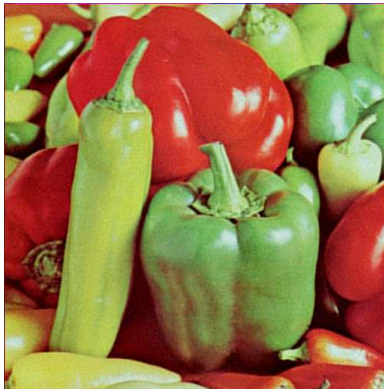


Error perception

Noise on high spatial frequencies



MSE: 16



SSIM: 0.882



Error perception

Chroma subsampling



MSE: 21.27

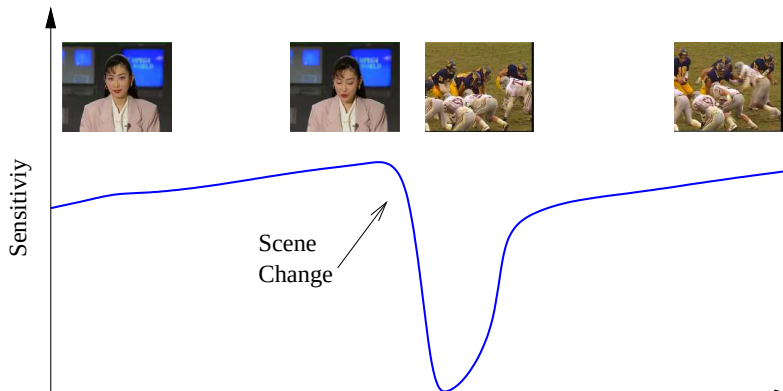


SSIM: —



Video Perception

- ▶ Sensitivity spatio-temporal frequencies
- ▶ Spatial and time masking





Perception and quality: summary

- ▶ Perceptual models are needed in order to achieve good compression performance
- ▶ Auditive system is relatively well understood and exploited in audio coders (see later on)
- ▶ HVS less well understood
- ▶ No P-OC totally reliable
- ▶ Nevertheless, the best performance can be achieved only taking into account HVS
- ▶ **Conclusions:** Modern approach consists in providing PSNR (simple, pixel-based), SSIM (structure-based) and LPIPS (feature-based) metric



Complexity, delay and robustness

- ▶ The complexity of a compression algorithm may be limited by:
 - ▶ Real-time constraints
 - ▶ Hardware limitations
 - ▶ Economical cost
- ▶ Delay is usually measured at the encoder
 - ▶ It is related to complexity ...
 - ▶ ... but mainly affected by the *coding order*
- ▶ Robustness: sensitivity of the compressed stream to losses



Performance criteria: summary

Contrasting issues:

↑↑ Quality	↓↓ Rate
↑↑ Robustness	↓↓ Complexity
	↓↓ Delay



Image Quality Comparison: Original





Image Quality Comparison: JPEG (1991)

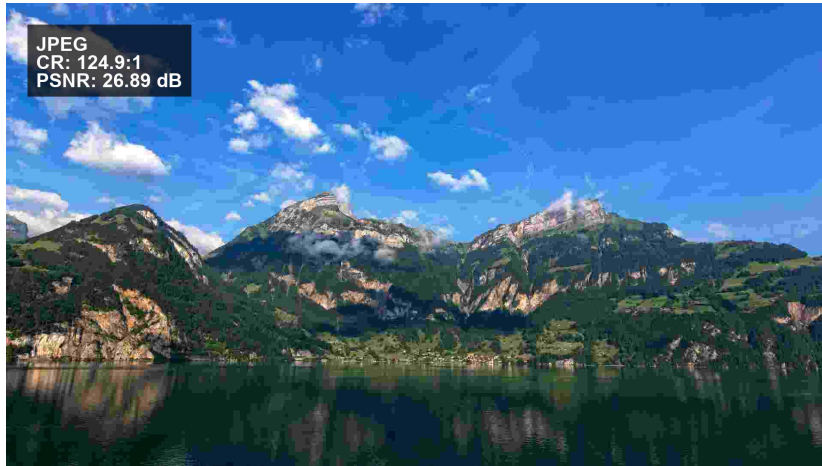




Image Quality Comparison: JPEG2000

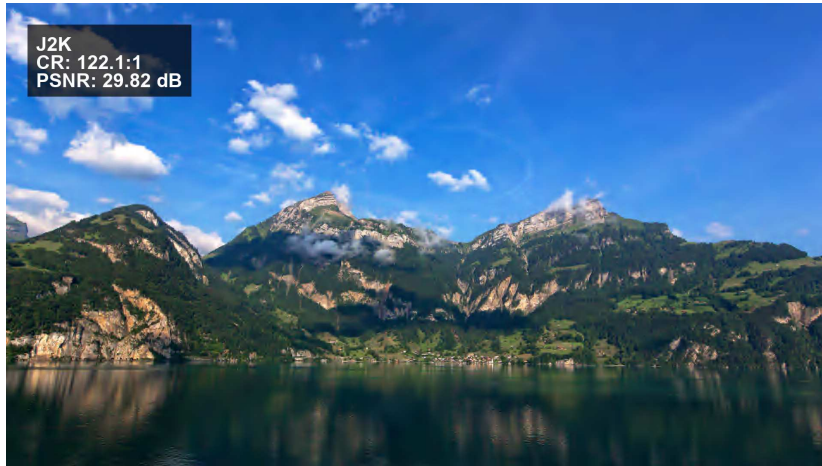




Image Quality Comparison: WebP (2010)

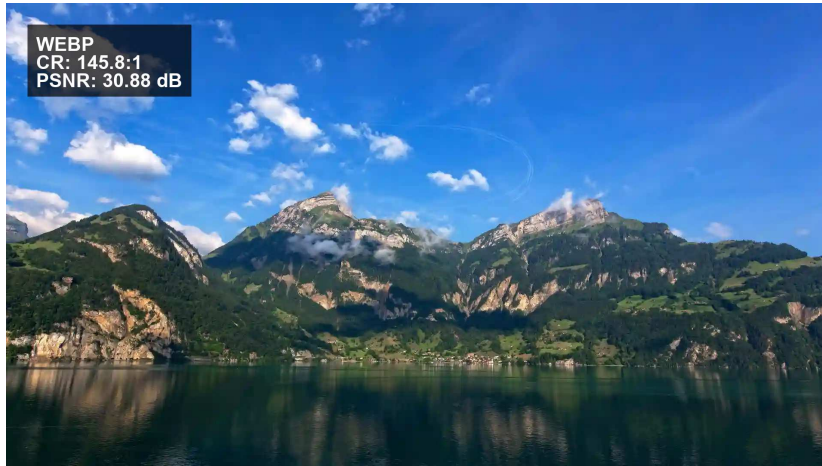




Image Quality Comparison: AVIF (2019)

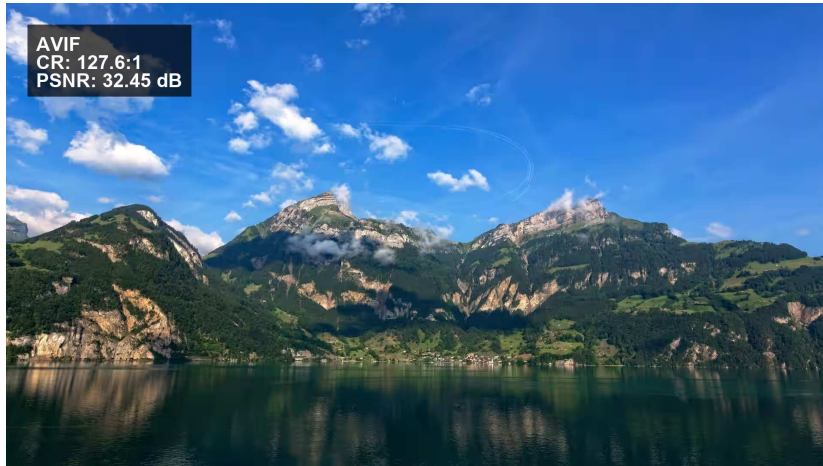




Image Quality Comparison: Color subsampling

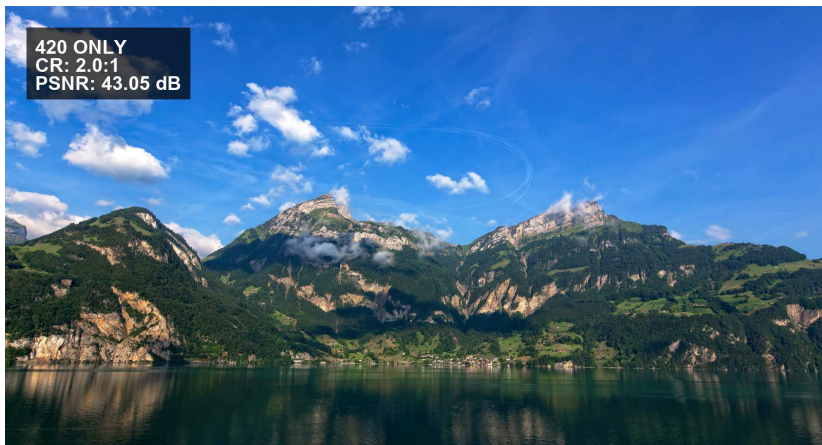
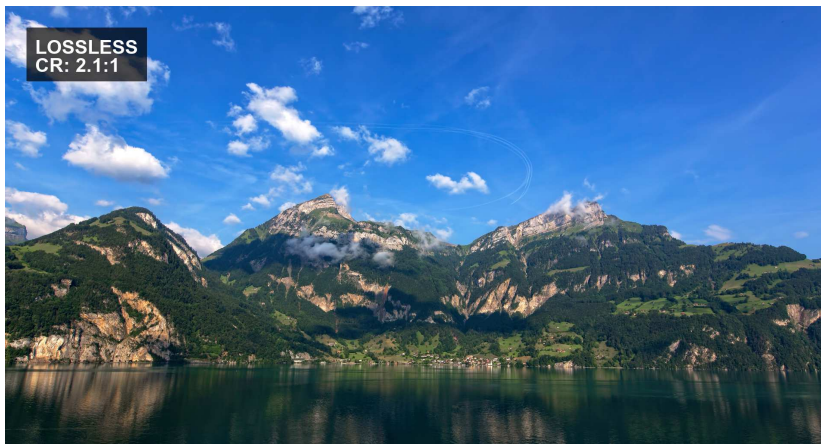




Image Quality Comparison: Lossless (PNG, 1996)





Compression Performance Summary

Table: Comparison of different codecs at a target budget of 50 KB

Codec	Size [Bytes]	Compression Ratio	PSNR [dB]
ORIGINAL	6220800	1.0:1	—
LOSSLESS	2923960	2.1:1	—
420 ONLY	3110400	2.0:1	43.05
JPEG	49795	124.9:1	26.89
J2K	50943	122.1:1	29.82
WEBP	42676	145.8:1	30.88
AVIF	48745	127.6:1	32.45